

Obesity in Indian children: Time trends and relationship with hypertension

MANU RAJ, K. R. SUNDARAM, MARY PAUL, A. S. DEEPA, R. KRISHNA KUMAR

ABSTRACT

Background. Limited data are available from India regarding the distribution and profile of childhood obesity and hypertension. We examined the time trends in childhood obesity in a representative sample of school children from Ernakulam District, Kerala and determined the relationship of obesity with blood pressure.

Methods. We used a stratified random cluster sampling method to select the children. Anthropometric data were collected from 24 842 students, 5–16 years of age, during 2003–04. Blood pressure and anthropometric data were collected from 20 263 students during 2005–06. Overweight and obesity were defined by body mass index for gender and age. Gender, age and height were considered for determining hypertension.

Results. The proportion of overweight children increased from 4.94% of the total students in 2003 to 6.57% in 2005 (OR: 1.36; 95% CI: 1.25–1.47; $p < 0.0001$). The increase was significant in both boys and girls. The proportion of overweight children was significantly higher in urban regions and in private schools, and the rising trend was limited to private schools. Systolic or diastolic incident hypertension was found in 17.34% of overweight children versus 10.1% of the remaining students (OR: 1.87; 95% CI: 1.60–2.17; $p < 0.0001$).

Conclusion. Childhood obesity showed an increasing trend in a short period of 2 years. Hypertension was common in overweight children. The results suggest the need for greater public awareness and prevention programmes on childhood obesity and hypertension.

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INTRODUCTION

Childhood overweight and obesity are global problems that are on the rise.¹ Obesity in children appears to increase the risk of subsequent morbidity, whether or not obesity persists into adulthood.² Outcomes related to childhood obesity include hypertension, type 2 diabetes mellitus, dyslipidaemia, left ventricular hypertrophy, non-alcoholic steatohepatitis, obstructive sleep apnoea, and orthopaedic and psychosocial problems.^{3–5}

According to WHO, 22 million children (under 5 years of age)

are overweight.¹ Obesity is evolving as a major nutritional problem in developing countries, affecting a substantial number of adults and resulting in an increased burden of chronic disease.¹ In national surveys conducted in the USA from the 1960s to the 1990s, the prevalence of overweight in children increased from 5% to 11%.⁶

Studies on urban Indian school children from selected regions report a high prevalence of obese and overweight children.^{7–11} In addition, studies on Indian school children have also demonstrated that the prevalence of hypertension in overweight children is significantly higher than that among normal children.^{12–15}

Till date no nationally representative data have emerged from India, which makes it difficult to project the prevalence of obesity and overweight among children in India. We aimed to determine the time trends of childhood obesity and overweight in a large population of school children from Ernakulam, Kerala, over a period of 2 years. The relationship of obesity with childhood hypertension was also explored.

METHODS

A contiguous area with a population of approximately 1.37 million was selected from Ernakulam district, in central Kerala. In the list of schools obtained from the District Education Office, Ernakulam, there were 377 schools with a total student strength of 202 710 in the area. Sampling was done by the stratified random cluster sampling method. The primary component of the study necessitated a sample size of 25 000. Schools were stratified into 5 groups according to the strength of children and a representative sample of 46 schools with a cumulative population of 25 228 children was randomly chosen. Consent to conduct the school survey and blood pressure measurement of the students was obtained from parents through school authorities who arranged parent meetings in the respective schools. Verbal assent was taken from the children after demonstrating and explaining the procedure.

Anthropometric data (height and weight) were collected from 24 842 students, 5–16 years of age, during 2003–04. Blood pressure and anthropometric data were collected from 20 263 students, 5–16 years of age, during the period 2005–06. Children with a body mass index (BMI) ≥ 85 th percentile of reference data were considered overweight and those with a BMI ≥ 95 th percentile were considered obese.³ The reference data used to identify the cut-off points were taken from the CDC 2000 dataset for BMI.¹⁶ Blood pressure (BP) was measured using standard methodology as recommended by The Fourth Report on the diagnosis, evaluation and treatment of high blood pressure in children and adolescents.¹⁷ Average systolic or diastolic BP ≥ 95 th percentile for gender, age and height was considered as hypertension. Pre-hypertension was defined as average systolic BP or diastolic BP that was ≥ 90 th

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percentile but <95th percentile. Children with BP levels ≥ 120 mmHg systolic and/or 80 mmHg diastolic were also considered pre-hypertensive.¹⁷

Children from the representative sample were called for screening according to their classes and were given rest for 5 minutes. The procedures were explained briefly and demonstrated to them. Those children who were cooperative and relaxed underwent BP measurement. Others were given adequate time to come to terms with the procedures. The BP was measured using a standardized mercury sphygmomanometer and recorded by trained paramedical personnel. BP was measured in a sitting posture with the hands resting on the examining table with the cubital fossa supported at the level of the heart. Chairs of adequate height were used for various groups. The stethoscope was placed over the brachial artery pulse, proximal and medial to the cubital fossa and below the bottom edge of the cuff (i.e. about 2 cm above the cubital fossa). Cuffs having a bladder width approximately 40% of the arm circumference midway between the olecranon and the acromion were used. The BP measurement was done on the right arm for consistency and comparison with standard tables. Three readings of the BP of each child were taken, maintaining an interval of 2 minutes between readings. The mean of 3 readings was reported. The weight and height of each child were recorded. Height was measured by a WHO-approved wall-mounted height measuring scale. A calibrated and standardized mechanical weighing scale was used to measure weight.

Statistical analysis

Statistical significance of the difference in the mean value of different variables between the 2 groups was tested by applying Student 't' test. Odds ratio and the corresponding confidence intervals were computed by using standard methods.¹⁸ The SPSS (version 11.0) software was used for this purpose.

RESULTS

In 2003, a total of 24 842 children (11 327 boys and 13 515 girls) and in 2005, 20 263 children (9754 boys and 10 509 girls) were examined. The descriptive data of both surveys are shown in Tables I and II. Overweight (including obesity) was found in 4.94% of the total students in 2003 and 6.57% in 2005. This increase is statistically significant (OR: 1.36; 95% CI: 1.25–1.47; $p < 0.0001$) and was seen in both sexes (Table III). A comparison of mean BMI between 2003 and 2005 showed an increase across all age groups. This increasing trend was seen in both boys and girls (Fig. 1). The prevalence of overweight was more in the age group of 5–11 years when compared with those in the age group of 12–16 years in 2003 and 2005. The difference in prevalence of overweight between the age groups was not statistically significant in 2003 (5.08% v. 4.78%, $p = 0.29$) but was significant in 2005 (7.08% v. 5.98%, $p < 0.005$).

Comparisons were made between overweight percentages in government and private schools (Table IV). Private schools had a higher percentage of overweight children than government schools, both in the 2003 (5.17% v. 3.83%; OR: 1.37; 95% CI: 1.15–1.62; $p < 0.0003$) and 2005 datasets (7.17% v. 3.23%; OR: 2.31; 95% CI: 1.87–2.86; $p < 0.0001$). An increasing trend was apparent among private schools (OR: 1.42; 95% CI: 1.30–1.54; $p < 0.0001$) and not among government schools (OR: 0.84; 95% CI: 0.65–1.09; $p > 0.05$). A significantly higher proportion of children from urban schools were obese compared with rural schools both in 2003 and 2005. Both urban and rural schools showed an increasing trend between 2003 and 2005 (Table V).

Blood pressure was examined in 20 263 children in the survey done during 2005. Incident hypertension was seen in 10.58% (7.17% in boys, 13.74% in girls) of total children. Systolic

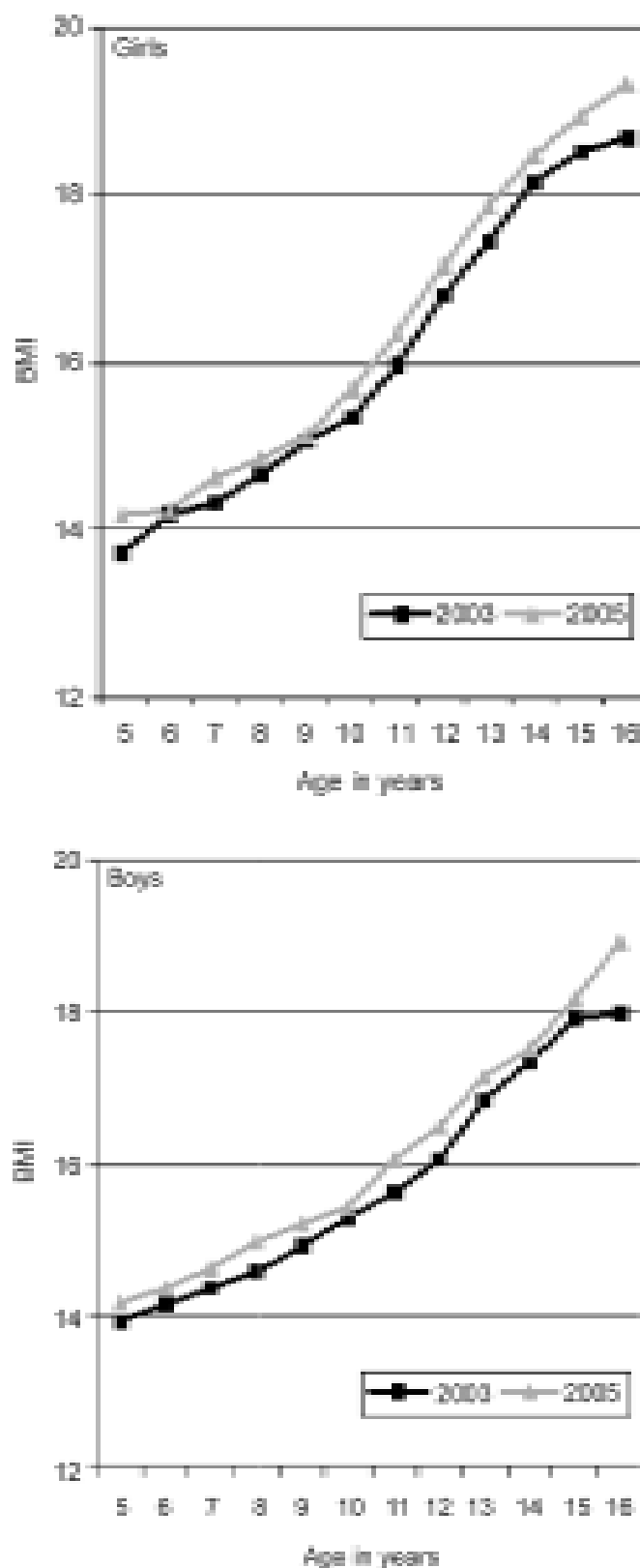


FIG 1. Body mass index (BMI) comparisons for 2003 and 2005 among girls and boys

TABLE I. Descriptive data of school survey, 2003

Age (years)	Boys				Girls			
	<i>n</i>	Height (cm)	Weight (kg)	BMI	<i>n</i>	Height (cm)	Weight (kg)	BMI
5	430	109.6 (6.23)	16.8 (2.86)	13.9 (1.38)	344	108.4 (5.33)	16.2 (2.29)	13.7 (1.50)
6	830	115.2 (5.59)	18.8 (3.27)	14.1 (1.68)	738	114.5 (6.04)	18.7 (3.53)	14.2 (1.79)
7	872	120.8 (6.17)	21.1 (4.11)	14.4 (1.88)	699	119.8 (6.08)	20.7 (3.91)	14.3 (1.81)
8	910	125.8 (6.19)	23.2 (4.38)	14.6 (1.84)	698	125.0 (6.42)	23.1 (4.98)	14.7 (2.12)
9	1016	131.2 (6.55)	25.9 (5.70)	14.9 (2.20)	1043	130.9 (6.55)	26.0 (5.59)	15.1 (2.31)
10	1235	135.5 (6.75)	28.3 (6.92)	15.3 (2.68)	1406	136.3 (7.09)	28.7 (6.33)	15.3 (2.45)
11	1213	140.4 (7.52)	31.0 (7.37)	15.6 (2.67)	1550	142.0 (7.47)	32.5 (7.36)	16.0 (2.72)
12	1340	145.7 (7.86)	34.4 (8.13)	16.1 (2.74)	1882	147.8 (7.14)	36.9 (7.95)	16.8 (2.80)
13	1266	152.8 (8.74)	39.7 (9.50)	16.8 (2.93)	2004	151.0 (6.57)	40.0 (7.59)	17.5 (2.73)
14	1282	158.6 (8.58)	44.0 (9.82)	17.3 (2.88)	1881	153.4 (6.05)	42.8 (7.79)	18.2 (2.83)
15	681	162.7 (8.04)	47.7 (9.71)	17.9 (2.80)	943	153.8 (5.80)	43.9 (7.55)	18.5 (2.88)
16	252	164.8 (7.43)	49.0 (9.12)	18.0 (2.56)	327	153.6 (6.17)	44.2 (7.75)	18.7 (2.93)

All values are mean (SD) BMI body mass index

TABLE II. Descriptive data of school survey, 2005

Age (years)	Boys						Girls					
	<i>n</i>	Height (cm)	Weight (kg)	BMI	SBP	DBP	<i>n</i>	Height (cm)	Weight (kg)	BMI	SBP	DBP
5	198	111.6 (5.63)	17.8 (3.19)	14.2 (1.53)	95.2 (8.15)	61.0 (8.41)	222	110.0 (5.52)	17.2 (2.89)	14.2 (1.57)	94.1 (9.25)	5.92 (9.20)
6	689	116.1 (5.61)	19.5 (3.77)	14.4 (1.85)	96.5 (8.68)	60.9 (8.93)	563	114.8 (5.55)	18.9 (3.36)	14.2 (1.66)	95.9 (8.44)	61.7 (7.97)
7	729	122.0 (5.61)	21.9 (4.71)	14.6 (2.20)	97.7 (8.22)	62.9 (8.75)	594	121.6 (5.73)	21.8 (4.61)	14.6 (2.20)	97.9 (8.35)	63.3 (8.32)
8	788	127.4 (6.26)	24.5 (5.32)	15.0 (2.26)	99.5 (9.09)	64.3 (8.68)	667	126.2 (6.33)	23.8 (4.99)	14.8 (2.16)	98.8 (9.36)	63.7 (8.51)
9	912	132.1 (6.32)	26.8 (5.76)	15.2 (2.32)	100.5 (8.56)	66.0 (8.31)	855	132.3 (6.58)	26.7 (5.69)	15.1 (2.32)	101.5 (9.36)	66.5 (8.38)
10	1109	137.1 (6.48)	29.3 (6.50)	15.4 (2.48)	102.1 (8.67)	67.3 (8.08)	1178	137.6 (7.08)	29.9 (7.10)	15.7 (2.62)	104.4 (9.87)	68.5 (8.17)
11	1126	141.9 (7.24)	32.6 (7.94)	16.1 (2.83)	103.5 (9.60)	68.3 (8.24)	1301	142.8 (7.33)	33.7 (8.01)	16.3 (2.77)	107.3 (10.01)	70.0 (7.85)
12	1125	146.7 (7.87)	35.9 (9.13)	16.5 (2.96)	105.3 (10.00)	68.6 (8.08)	1269	148.6 (6.68)	38.2 (8.07)	17.2 (2.83)	109.8 (10.35)	71.7 (7.62)
13	1024	153.7 (8.83)	40.9 (9.71)	17.2 (2.99)	108.0 (11.20)	69.1 (8.67)	1388	152.1 (6.24)	41.5 (7.82)	17.9 (2.84)	112.3 (10.57)	72.6 (7.80)
14	1088	159.8 (8.36)	45.1 (9.60)	17.5 (2.79)	111.0 (11.29)	71.2 (8.25)	1481	154.2 (5.99)	44.0 (8.53)	18.5 (3.08)	113.2 (10.42)	73.3 (7.85)
15	687	164.2 (7.71)	49.3 (10.51)	18.2 (3.05)	113.8 (10.92)	72.8 (8.46)	813	155.3 (6.28)	45.8 (8.39)	19.0 (2.88)	114.4 (10.47)	74.2 (7.94)
16	279	165.9 (7.50)	52.4 (11.42)	18.9 (3.28)	115.1 (11.44)	73.2 (8.20)	178	155.2 (6.58)	46.6 (8.92)	19.3 (3.28)	114.7 (10.87)	74.5 (7.10)

BMI body mass index SBP systolic blood pressure in mmHg DBP diastolic blood pressure in mmHg All values are mean (SD)

TABLE III. Percentages of overweight and obese children in 2005 and 2003

Dataset	Overweight			Obese		
	Overall	Boys	Girls	Overall	Boys	Girls
2005 (<i>n</i> =20 263)	6.57	7.33	5.87	1.89	2.47	1.34
2003 (<i>n</i> =24 842)	4.94	5.38	4.57	1.26	1.65	0.93
Odds ratio	1.36*	1.39	1.30	1.51*	1.51	1.45
95% confidence interval	1.25–1.47	1.24–1.56	1.16–1.46	1.29–1.76	1.24–1.84	1.13–1.85
p value	<0.0001	<0.0001	<0.0001	<0.0001	<0.0001	<0.003

* Odds ratio adjusted for gender and age

TABLE IV. Percentages and trends of overweight children in private and government schools

Dataset	Private schools	Government schools	Odds ratio, 95% CI, p value
2005	7.17	3.23	2.31, 1.87–2.86, <i>p</i> <0.0001
2003	5.17	3.83	1.37, 1.15–1.62, <i>p</i> <0.0003
Odds ratio	1.42*	0.84*	
95% CI	1.30–1.54	0.65–1.09	
p value	<i>p</i> <0.0001	<i>p</i> >0.05	

* Odds ratio adjusted for gender and age CI confidence interval

TABLE V. Percentages and trends of overweight children in urban and rural schools

Dataset	Urban	Rural	Odds ratio, 95% CI, p value
2005	8.66	3.75	2.43, 2.13–2.77, <i>p</i> <0.0001
2003	6.43	2.91	2.29, 2.00–2.62, <i>p</i> <0.0001
Odds ratio	1.38*	1.30*	
95% CI	1.26–1.52	1.11–1.53	
p value	<i>p</i> <0.0001	<i>p</i> <0.005	

* Odds ratio adjusted for gender and age CI confidence interval

hypertension was seen in 5.84% (3.20% in boys, 8.29% in girls) of children and diastolic hypertension in 6.61% (4.89% in boys, 8.2% in girls). Among the total children, 10.65% (9.33% in boys, 11.89% in girls) had systolic pre-hypertension and 14.75% (12.53% in boys, 16.8% in girls) had diastolic pre-hypertension.

Systolic BP correlated better with weight (boys $r=0.559$, girls $r=0.539$) and diastolic BP with height (boys $r=0.399$, girls $r=0.458$). The correlation of BP with BMI was lower compared with correlation of BP with weight as well as height in both sexes. In both boys and girls, the mean systolic BP in the overweight group was more than that in the normal weight group. In boys, the mean diastolic BP in the overweight group was more than that in the normal weight group in all ages except in those who were 15 years old. In girls, the mean diastolic BP in the overweight group was more than that in the normal weight group in all age groups (Table VI).

The prevalence of incident hypertension in normal weight, overweight and obese groups was 10.1%, 17.34% and 18.32%, respectively. In the total sample, children in the age group of 12–16 years had a higher prevalence of incident hypertension when

compared with those in the age group of 5–11 years (13.18% v. 8.35%). The prevalence of systolic hypertension (first instance) in normal weight, overweight and obese groups was 5.38%, 12.31% and 14.66%, respectively. The prevalence of diastolic hypertension (first instance) in normal weight, overweight and obese groups was 6.45%, 8.86% and 8.9%, respectively. The sex-wise differences in various blood pressures among the 3 weight groups are given in Table VII.

DISCUSSION

Although the prevalence values of childhood obesity in this study are lower than those of other studies from similar settings,^{7–11} the increasing trend of overweight and obesity in our study was significant. During a short span of 2 years the proportion of overweight children increased significantly across all age groups and in both sexes. This was accompanied by an increasing trend in mean BMI values across all age groups. The increase was substantial in private schools and not seen in government schools. Although urban schools had a higher proportion of overweight children, both urban and rural schools showed an upward trend in

the 2-year period. This rapid increase in a short span of 2 years assumes considerable epidemiological importance. A high BMI correlates strongly with markers of secondary complications of obesity including current blood pressures,^{19–21} blood lipid and lipoprotein levels^{19,22–24} and long term mortality.²

Evidence from several national health surveys in Asia points to significant differences in prevalence of overweight and obesity among countries.^{25–29} Asian countries such as Taiwan and China have experienced rapid increases in prevalence of childhood obesity.^{30,31} Rapid economic growth has improved the nutritional, socioeconomic and health status of many countries.³² Obesity has increased markedly with this nutritional evolution in most Asian countries.³³ A similar nutritional transition is under way in India as well.²⁹ In addition to the nutritional and socioeconomic transitions, the behavioural transition of children is also possibly contributing significantly to the rapidly rising prevalence of obesity. Unhealthy eating habits and physical inactivity are the major culprits.³⁴ The sedentary lifestyle of children and adolescents have been attributed mainly to television viewing, computer games, internet, over-emphasis on academic excellence, unscientific urban planning

TABLE VI. Comparison of systolic blood pressure (SBP) and diastolic blood pressure (DBP) between normal and overweight groups

Age (years)	Group	Boys				Girls			
		SBP (mmHg)		DBP (mmHg)		SBP (mmHg)		DBP (mmHg)	
		Mean (SD)	p value	Mean (SD)	p value	Mean (SD)	p value	Mean (SD)	p value
5	NW	95.1 (8.25)	0.295	60.8 (8.43)	0.065	93.7 (9.10)	0.004	58.8 (9.28)	0.001
	OW*	97.9 (5.84)		65.8 (6.90)		101.6 (9.19)		65.1 (4.98)	
6	NW	96.1 (8.55)	0.001	60.5 (8.88)	0.001	95.8 (8.37)	0.041	61.5 (7.94)	0.023
	OW	102.0 (8.72)		66.1 (8.00)		99.1 (9.28)		65.0 (8.04)	
7	NW	97.1 (8.02)	0.001	62.5 (8.61)	0.001	97.6 (8.30)	0.001	62.8 (8.33)	0.001
	OW	103.8 (7.86)		66.6 (9.26)		102.3 (7.86)		68.7 (5.90)	
8	NW	99.0 (9.01)	0.001	63.9 (8.60)	0.001	98.4 (9.22)	0.001	63.7 (8.56)	0.512
	OW	104.2 (8.53)		67.7 (8.81)		104.5 (9.73)		64.6 (7.69)	
9	NW	99.9 (8.34)	0.001	65.6 (8.30)	0.001	101.2 (9.28)	0.001	66.2 (8.44)	0.001
	OW	107.7 (7.64)		69.8 (7.47)		107.4 (8.89)		70.3 (6.37)	
10	NW	101.6 (8.62)	0.001	67.0 (8.10)	0.001	103.9 (9.73)	0.001	68.3 (8.09)	0.001
	OW	108.2 (6.90)		70.5 (7.03)		111.2 (9.69)		71.9 (8.60)	
11	NW	102.8 (9.42)	0.001	67.8 (8.21)	0.001	106.9 (9.94)	0.001	69.8 (7.78)	0.001
	OW	110.5 (8.60)		72.6 (7.35)		111.9 (9.85)		72.7 (8.28)	
12	NW	104.7 (9.84)	0.001	68.5 (8.03)	0.026	109.5 (10.38)	0.001	71.7 (7.62)	0.095
	OW	111.6 (9.73)		70.5 (8.41)		114.8 (8.45)		73.2 (7.59)	
13	NW	107.5 (11.00)	0.001	68.9 (8.67)	0.035	112.0 (10.47)	0.001	72.6 (7.79)	0.509
	OW	114.3 (12.09)		71.2 (8.52)		117.4 (10.78)		73.2 (8.03)	
14	NW	110.6 (11.14)	0.001	71.0 (8.20)	0.002	112.9 (10.30)	0.001	73.1 (7.88)	0.002
	OW	119.7 (10.83)		74.7 (8.71)		119.3 (10.62)		75.9 (6.97)	
15	NW	113.4 (10.94)	0.001	72.8 (8.52)	0.984	114.0 (10.28)	0.001	74.1 (7.86)	0.010
	OW	119.6 (9.03)		72.7 (7.55)		121.3 (11.92)		77.4 (8.96)	
16	NW	114.6 (11.33)	0.002	73.0 (8.21)	0.169	114.2 (10.58)	0.008	74.3 (7.21)	0.026
	OW	124.4 (9.74)		76.1 (7.77)		123.9 (12.80)		77.6 (3.45)	

*Overweight includes obese children NW normal weight group (children who are neither overweight nor obese) OW overweight group

TABLE VII. Percentages of different forms of hypertension in children with respect to gender and weight group

Group	Normal (n=18 931)			Overweight (n=1332)			Obese (n=382)		
	Boys	Girls	Total	Boys	Girls	Total	Boys	Girls	Total
Hypertension	6.76	13.15	10.10	12.31	23.18	17.34	14.94	24.11	18.32
Systolic hypertension	2.82	7.72	5.38	7.97	17.34	12.31	10.79	21.28	14.66
Systolic pre-hypertension	8.76	11.62	10.25	16.50	16.21	16.37	18.67	23.40	20.42
Diastolic hypertension	4.74	8.02	6.45	6.85	11.18	8.86	7.88	10.64	8.90
Diastolic pre-hypertension	12.16	16.46	14.40	17.20	22.37	19.59	17.84	26.24	20.94

and ever-increasing automated transport.³⁴ The difference in prevalence and trends of overweight among rural and urban schools demonstrated by the study could be due to these influences. However, it is interesting that there was an upward trend noticeable in rural schools as well. Lifestyle changes are likely to have a greater impact on children from affluent backgrounds as suggested by the observation that the rapid rise in prevalence rates of overweight was largely limited to private schools.

The prevalence of incident hypertension in normal weight children in our study was 10.1%. The study limitations did not permit us to follow up cases of incident hypertension and find out actual values of persistent hypertension which would have been much smaller than the quoted figure. This phenomenon of low prevalence of hypertension on subsequent readings has been demonstrated previously. Sorof *et al.*³⁵ documented first instance hypertension as 19%, second instance as 9.5% and third instance (persistent) as 4.5%.

The relationship of overweight and hypertension was examined in detail. Hypertension was seen in significantly higher percentages among overweight children when compared with normal children (17.34% v. 10.1%). This relationship has been reported by other studies as well.^{12,36-41} Rosner *et al.*⁴² pooled data from 8 large epidemiological studies. Irrespective of race, gender or age, the risk of elevated BP was significantly higher for children in the upper compared with the lower decile of BMI, with an odds ratio of systolic hypertension ranging from 2.5 to 3.7. Freedman *et al.*³⁹ reported that overweight children in the Bogalusa heart study were 4.5 and 2.4 times as likely to have elevated systolic BP and diastolic BP, respectively. Studies from India have also shown similar trends.¹²⁻¹⁵

Evidence from prospective studies in Asia suggests that obesity is directly related to the incidence of hypertension, type 2 diabetes and hypercholesterolaemia.^{26,43} According to WHO, chronic diseases are the major causes of death in almost all countries including those in Asia.¹ It is estimated that 70% of these deaths will occur in developing Asian nations such as China, India, Pakistan, Cambodia and Viet Nam.¹ Cardiovascular diseases are responsible for the major share of these deaths due to chronic diseases.¹ South Asians appear to have worse cardiovascular disease risk profiles when compared with Caucasian populations with similar BMI levels.^{26,43} Small increases in BMI among them may translate into a substantial increase in the burden of cardiovascular diseases. For this reason the trends reported here are alarming even though the overall proportion of obesity and overweight children is far lower than that reported in other studies.

Childhood obesity and related hypertension are issues related to lifestyle patterns in the population. Even though the situation discussed is alarming, it provides an excellent opportunity for prevention of cardiovascular diseases by means of lifestyle interventions targeted at childhood and adolescent populations, thereby attempting to reduce morbidity and mortality arising from these diseases in the future.

Limitations of the study

Our study did not include measurements of waist-hip ratio and skin-fold thickness. BP readings in 2003 also would have been useful in understanding the trend. Application of international reference standards of BMI and hypertension in an Indian setting may have limitations. We measured incident hypertension; hypertension can only be confirmed after a minimum of 3 separate BP measurements demonstrating high BP. As BP was measured

in the field, an element of anxiety and apprehension might have affected a subset of children in our study. The data provided are univariate.

Conclusions

Childhood obesity showed a significantly increasing trend in a short period of 2 years. Hypertension was seen in high percentages in children with overweight and obesity when compared with children who were neither overweight nor obese. Unless effective interventions and preventive strategies are instituted at the local and national level, these observations suggest that the trend of increasing cardiovascular disease in adults observed in recent decades will accelerate even further. These trends are disturbing and call for concerted efforts targeted at improving lifestyles of children and adolescents.

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